Google Cloud H2S

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Participant Details

a. **Participant Name: Krithik Gokul S**
b. **Problem Statement:** Multi-Agent Productivity Assistant

Brief about the idea

A Multi-Agent AI Productivity Assistant that helps users manage tasks, schedules, and notes by coordinating multiple intelligent agents. The system integrates with real-world tools like calendars, task managers, and databases to automate workflows and improve productivity.

Your solution should be able to explain the following:

Approach & Solution:
We built a multi-agent system using ADK with a coordinator agent managing task, calendar, and notes agents. MCP connects external tools, and AlloyDB stores structured data, enabling end-to-end automated workflows via an API.
Real-World Problem & Impact:
Users struggle with managing tasks across multiple apps. Our solution centralizes everything, automates actions, and improves productivity by saving time and reducing manual effort.
Core Workflow:
User input → Coordinator agent → Sub-agent selection → MCP tool execution → Data stored/retrieved from DB → Final response returned.

Typeset

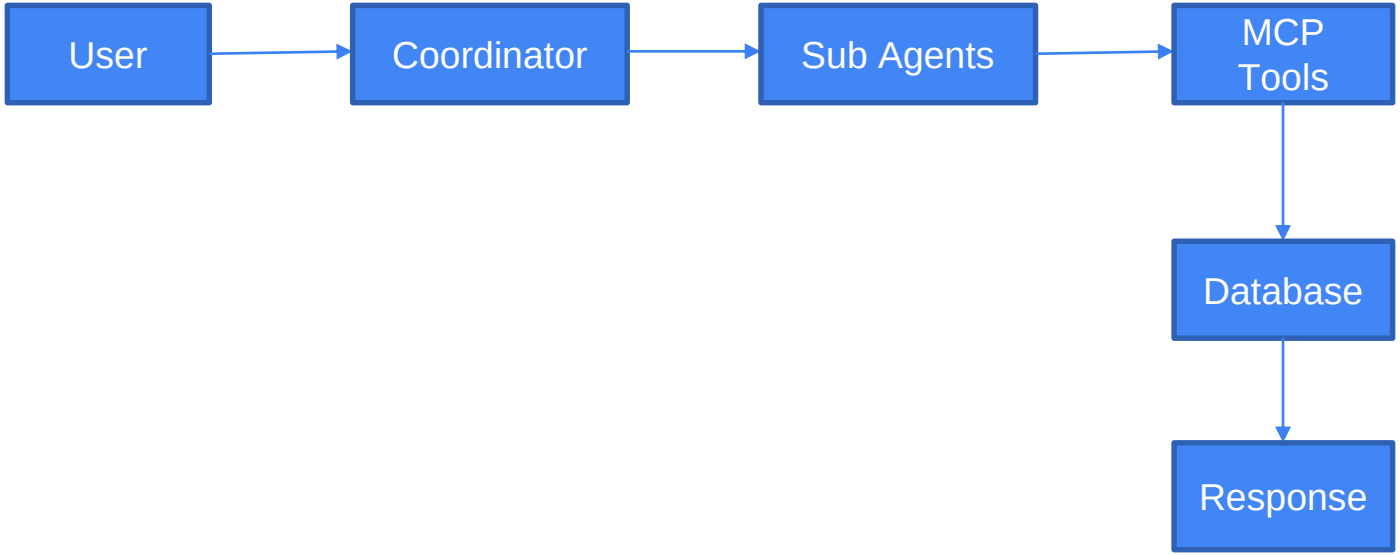
Opportunities

Difference from Existing Solutions:
Unlike traditional productivity tools, this system uses multiple AI agents that collaborate and automate tasks instead of relying on manual input.
USP (Unique Selling Point):
Multi-agent coordination with real-time tool integration (MCP) and persistent memory (AlloyDB), enabling intelligent, automated, and context-aware workflows.

List of features offered by the solution

- ✅ Task creation & tracking
- ✅ Smart scheduling
- ✅ Note summarization
- ✅ Multi-step workflow execution
- ✅ Context-aware reminders
- ✅ Natural language interaction
- ✅ Database-backed memory
- ✅ API integration

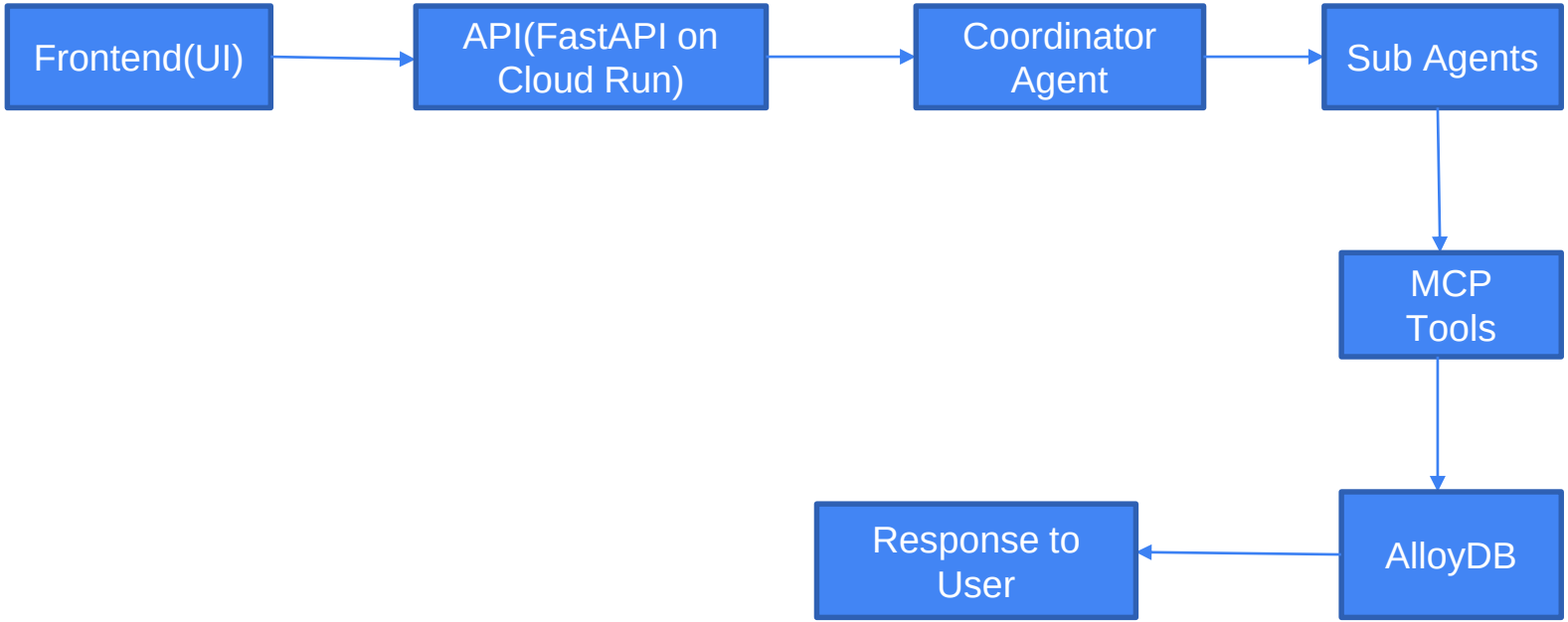
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution

Simple dashboard with task list, calendar view, and notes panel, along with a chat interface for user interaction and AI-driven actions

Architecture diagram of the proposed solution:



Technologies / Google Services:
ADK, Gemini API, MCP, AlloyDB, FastAPI, Cloud Run.
Why This Stack:
Enables scalable, API-based deployment with real-time tool integration and persistent data storage, supporting efficient multi-agent workflows in real-world applications.

Snapshots of the prototype

Google Cloud Credit and Accou | my-alloydb-cluster-1 – AlloyDB | AlloyDB Setup Manager | Ask Gemini

console.cloud.google.com/alloydb/locations/us-central1/clusters/my-alloydb-cluster-1/studio?cloudshell...

Google Cloud | My Project 89243 | Search (/) for resources, docs, products, and more | Search

All clusters > Primary cluster my-alloydb-cluster-1: AlloyDB Studio > Primary instance my-primary-inst-1

Explorer

- postgres
  - Schema 7
    - public (Default)
      - Tables 3
      - Views 48
      - Functions 143
      - Procedures 0
      - Sequences 3
      - Types 0
      - Extensions 5
      - BigQuery Tables 0 ...
      - Foreign Tables 0 ...
    - ai
    - google_ml

Untitled query | Gemini settings

Run | Save | Format | Clear | Valid

```
CREATE TABLE tasks (
    id SERIAL PRIMARY KEY,
    content TEXT
);

CREATE TABLE calendar (
    id SERIAL PRIMARY KEY,
    content TEXT
);

CREATE TABLE notes (
    id SERIAL PRIMARY KEY,
    content TEXT
);
```

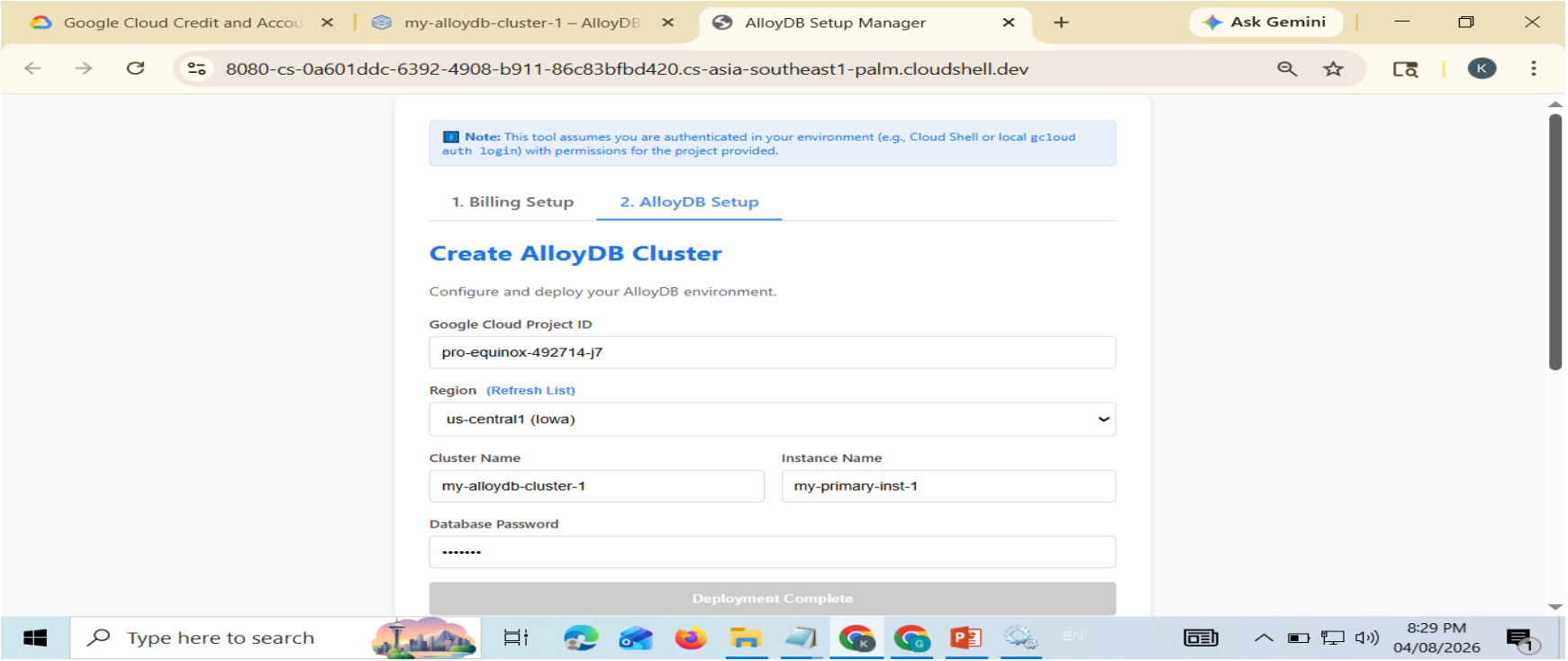
Results | Explanation | Execution time: 71.3 ms

Statement executed successfully

CLOUD SHELL Editor | Open Terminal

Type here to search | 8:29 PM 04/08/2026

Snapshots of the prototype



Snapshots of the prototype

Google Cloud Credit | my-alloydb-cluster-1 | 8000-cs-0a601ddc-6 | AlloyDB Setup Mana | Ask Gemini

8000-cs-0a601ddc-6392-4908-b911-86c83bfbd420.cs-asia-southeast1-palm.cloudshell.dev

Pretty-print


```
{
  "message": "Multi-Agent Productivity Assistant Running"
}
```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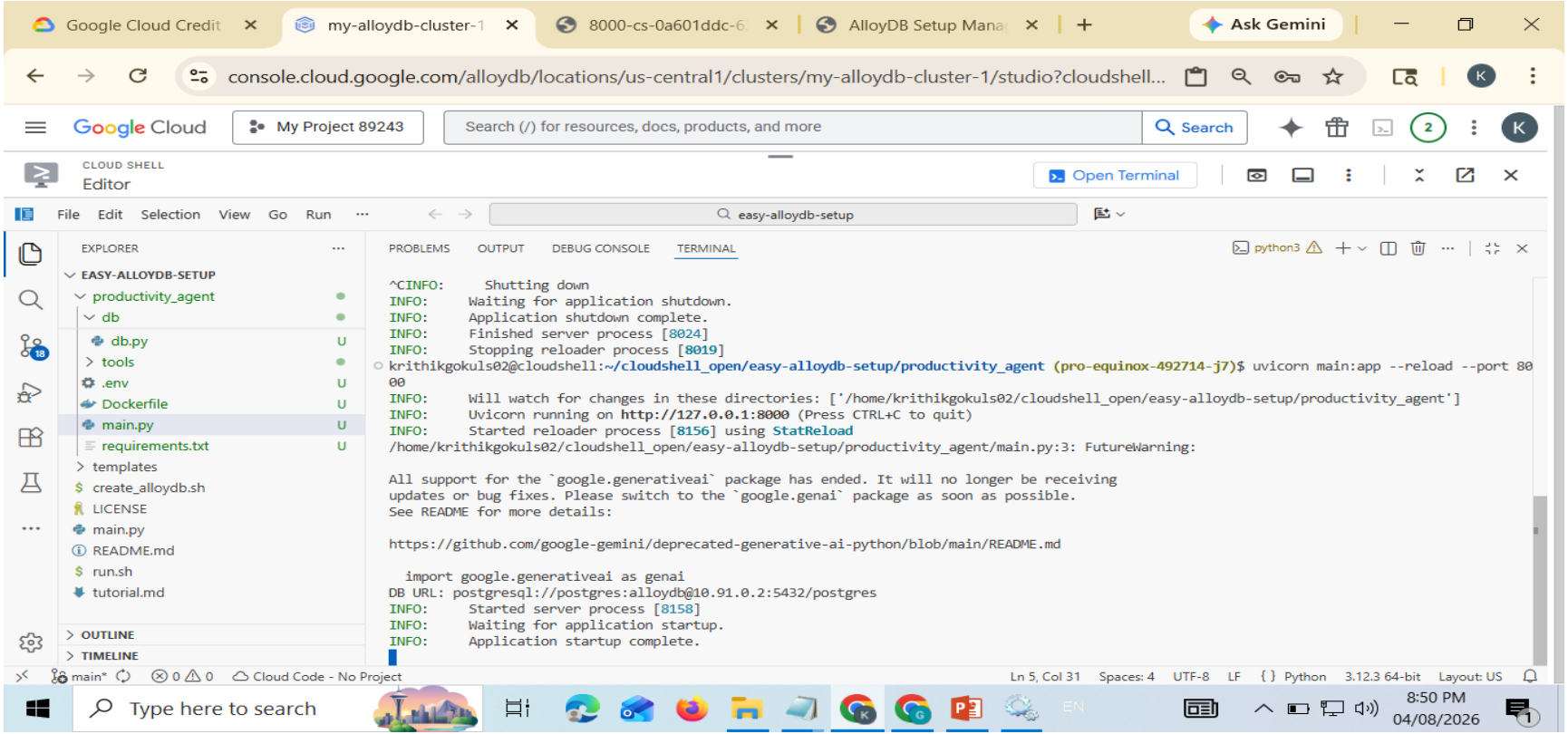
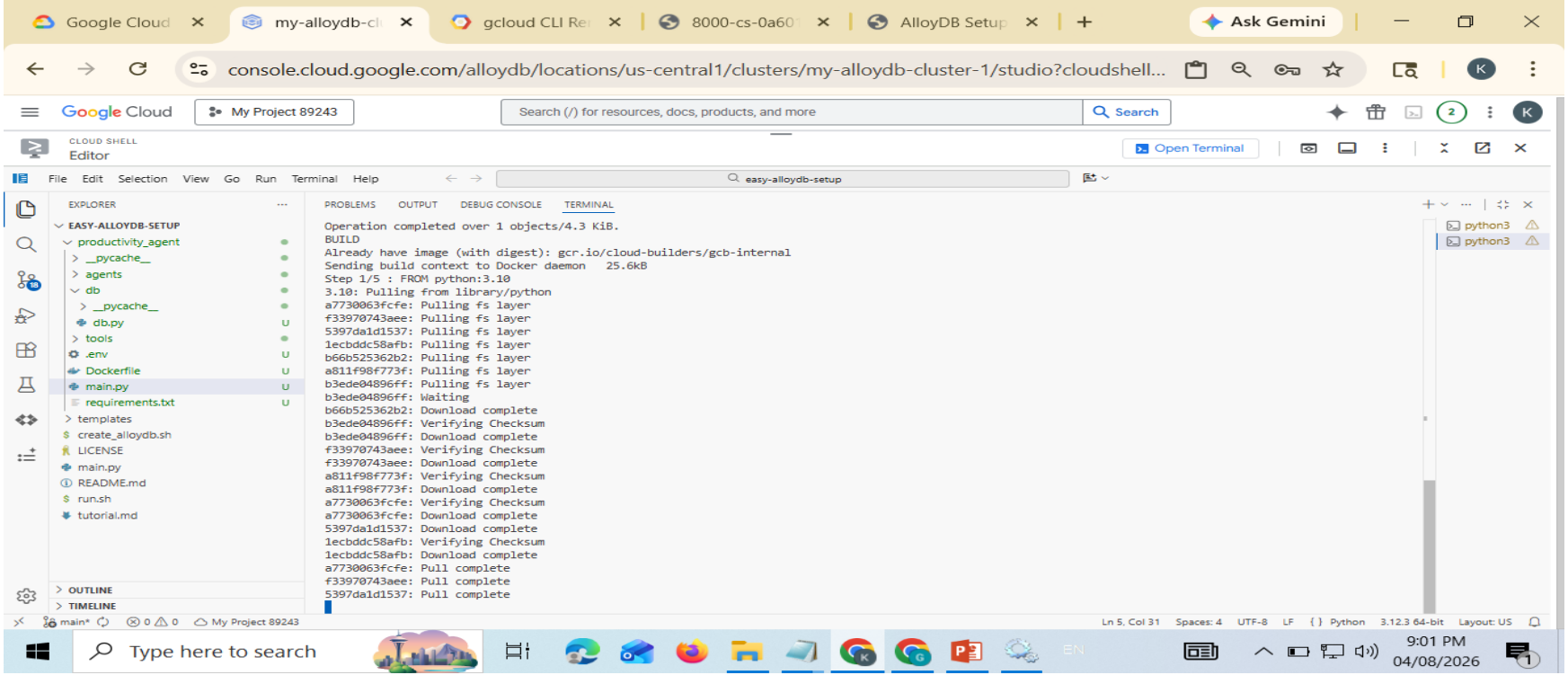
04/08/2026

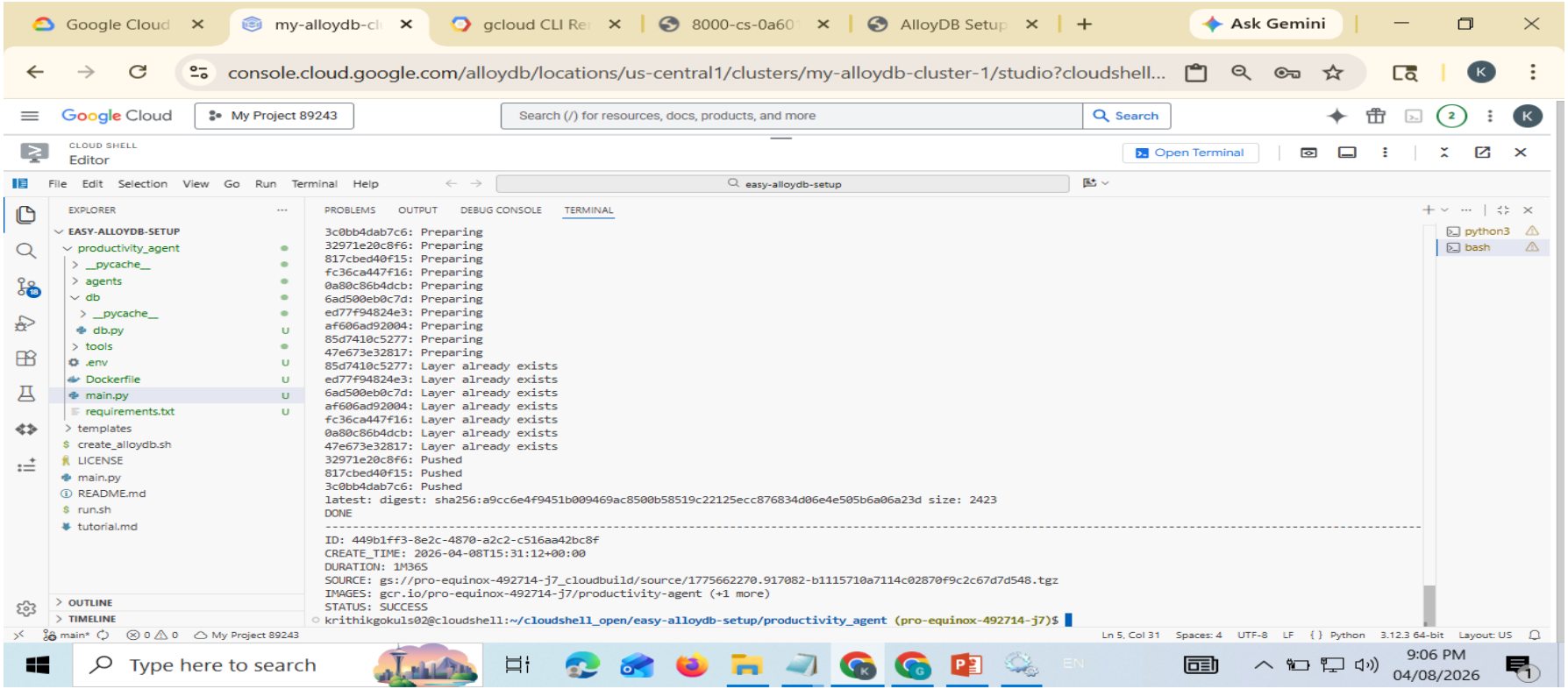
Snapshots of the prototype



Snapshots of the prototype



Snapshots of the prototype



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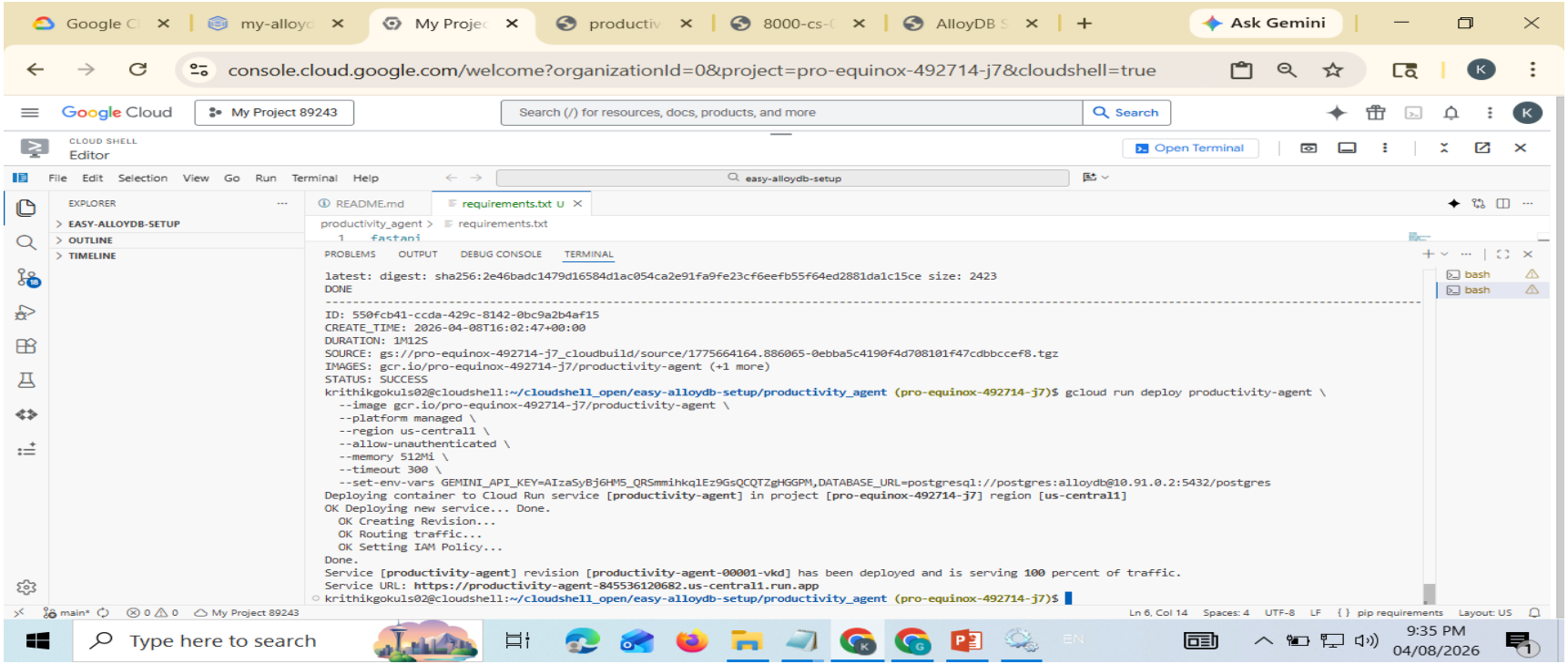

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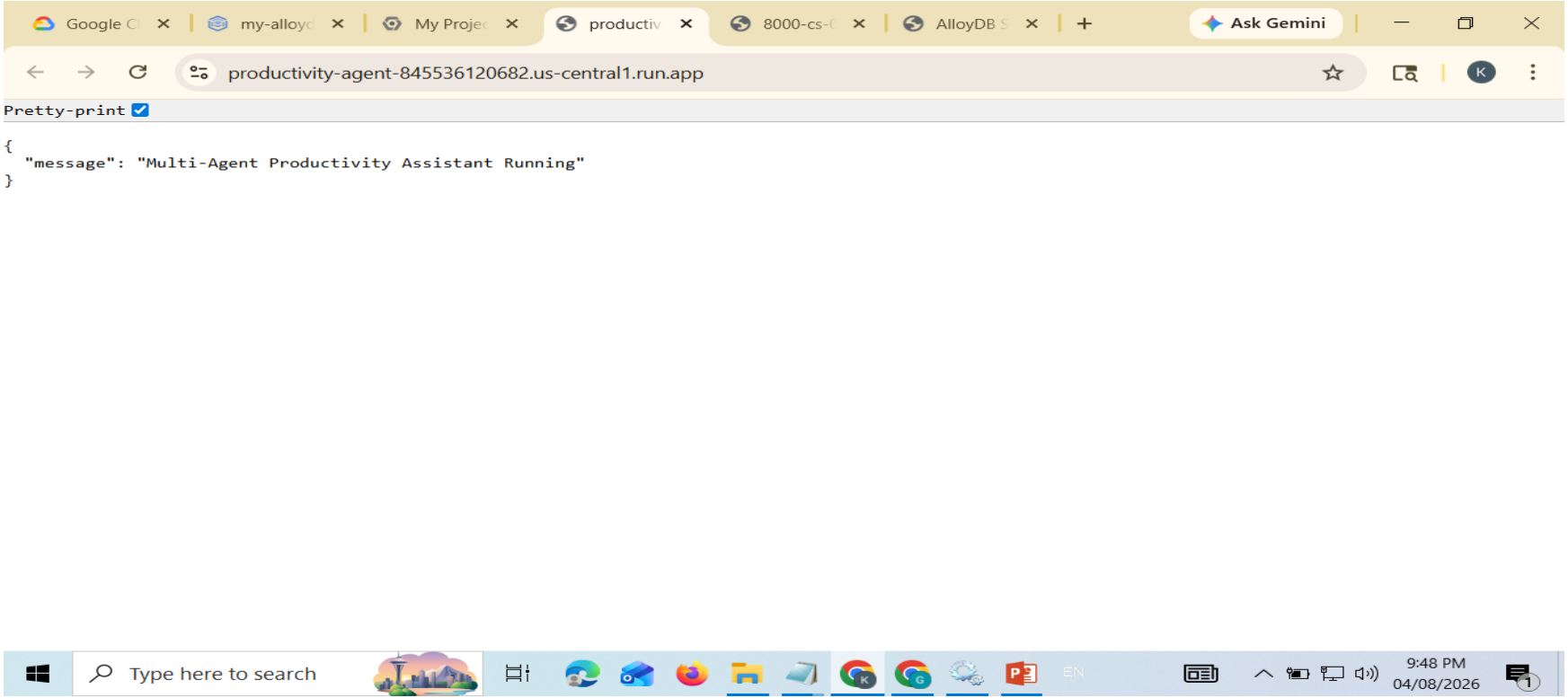

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Thank you